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THE COPPERBELT UNIVERSITY

SCHOOL OF MEDICINE

END OF TERM II EXAMINATIONS - JUNE 2019

MBS 210 and MBM 220

FOLLOWING INSTRUCTIONS IS PART OF EXAMINATION

PHYSIOLOGY

STUDENT NUMBER: 17113672 **PROGRAMME:** Bsc biomed

TIME ALLOWED: 2 HOURS

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INSTRUCTIONS:

1. Do not write your name/contact number or any sign of disclosure of identity on any page
2. Write your STUDENT NUMBER ONLY on every page of the answer sheet.
3. Any unclear mark/answer will be deemed incorrect.
4. Answer **ALL** questions in Sections A and B
5. Answer **Question ONE** (Compulsory) in **Section C** and **TWO other** questions. (3)
6. Answer **Section C** on the separate sheets attached, with each question on a new page

IMPORTANT

Success is the step-wise attainment of pre-defined goals in a deliberate, consistent and persistent manner. It begins with the discovery of purpose. Why are you studying what you are studying?

SECTION A: Multiple Choice Questions
Choose the best Option (1mark each)

1. A membrane transport protein has a "transporter" mechanism if:
   T A. it forms an open pore through which a molecule can diffuse.
   B. an electrochemical gradient is necessary for transport to occur.
   C. it only allows transport "down" a concentration gradient.
   D. molecules are transported in opposite directions across the membrane.
   E. it binds to the molecule and changes shape during transport.

2. In a living neuron, an action potential is triggered when:
   T A. an electrode raises the potential above the threshold level.
   B. potassium ions flow into the dendrite.
   F C. voltage-gated Ca - channels open in the post-synaptic membrane.
   F D. voltage-gated K - channels close along the axon.
   E. ligand-gated Na - channels open in the post-synaptic membrane.

3. A "passive" membrane transport protein:
   F A. will require a direct source of energy for the transport to occur.
   T B. can only transport a molecule 'down' a gradient.
   C. is most likely to involve a 'carrier' type transport mechanism.
   F D. can move a molecule 'up' a gradient if a membrane potential exists.

4. Suppose you were to treat a normal mammalian cell with a substance that inhibits the Na-K ATPase. What would be the most **immediate** effect upon the cell?
   F A. There would be no change at all.
   T B. The cell's osmotic balance would be disrupted, and the cell would begin to swell.
   C. The cell membrane potential would immediately drop to zero.
   D. The cell would very quickly run out of ATP.
   E. The cell could not create an action potential.

5. Voltage-gated Na+ Channels:
   A. Remain open until the membrane potential reaches the threshold value.
   B. First transport NA+ into the cell, and then change the direction of transport.
   T C. Spontaneously become inactivated a few microseconds after they open.
   D. Close, but remain active when an action potential is achieved.

6. For a typical animal cell, the membrane potential is determined by the equilibrium distribution o K+ ions across the cell membrane. Free movement of K+ across the membrane occurs through:
   F A. the voltage gated K-channel
   T B. the K+ leak channel
   F C. the Na-K pump
   T D. free diffusion

7. Transport across a membrane is said to be 'coupled' when
   T A. two molecules are transported across the membrane in the same direction.
   F B. membrane transport is coupled to an energy source, such as ATP hydrolysis.

T C. transport of one ion down its gradient provides the energy to transport another molecule against its gradient.

F D. both the concentration gradient and membrane potential determine the rate of transport across the membrane.

8. A membrane transporter is said to be 'gated' if it:
   T A. requires the binding of ATP to open.
   F B. allows molecules to pass in only one direction.
   F C. participates in the formation of the membrane potential
   T D. opens in response to a specific stimulus.

9. The membrane potential across the cell membrane is the result of:
   T A. the difference in concentration of K+ and Na+ on either side of the membrane.
   T B. more positively-charged ions outside and more negatively-charged ions inside. ✓
   T C. opening and closing of voltage-gated channels in the membrane.
   F D. electrodes that generate different charges on either side of the membrane.

10. A human white blood cell engulfs a bacterial cell by:
    F A. Carrier-mediated facilitated diffusion.
    B. Exocytosis.
    T C. Phagocytosis. ✓
    T D. Pinocytosis.
    F E. The sodium-potassium pump.

11. A bacterium containing sodium ions at a concentration of 0.1 mM lives in a pond that contains sodium ions at 0.005 mM. Evidently, sodium ions are entering the cell by:
    A. Active transport. ✓
    B. Endocytosis.
    C. Diffusion.
    D. Facilitated diffusion.
    false F E. Osmosis

12. Although glucose molecules constantly diffuse into a cell along their concentration gradient, equilibrium is never reached and glucose continues to enter the cell. This is a direct result of:
    A. The very fast turnover rate of glucose metabolism.
    B. The continuous excretion of glucose from other parts of the cell
    C. The rapid and continuous intracellular formation of glucose phosphates.
    D. The active transport of glucose.
    E. The ability of the cell to engulf glucose by pinocytosis.

13. Penicillin is toxic to certain dividing bacterial cells because it prevents cell wall formation, ca the cells to burst. This indicates that the bacteria live in:
    A. A hypotonic medium.
    B. A hypertonic medium.
    C. An isotonic medium.
    D. A medium with a higher osmotic pressure than the cell.
    E. Both a hypertonic medium and a medium with a higher osmotic pressure than the cel

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(RBC)

144 S'

(water)

14. A patient who has had a severe hemorrhage accidentally receives a large transfusion of distilled water directly into a major blood vessel. You would expect this mistake to:

A. Have no unfavorable effect as long as the water is free of bacteria.

B. Have serious, perhaps fatal consequences because there would be too much fluid to pump.

C. Have serious, perhaps fatal consequences because the red blood cells could shrink.

D. Have serious, perhaps fatal consequences because the red blood cells could swell and burst.

E. Have no serious effect because the kidney could quickly eliminate excess water.

15. Simple diffusion may involve the movement of ________________ through the plasma membrane down a concentration gradient.

T A. Small polar molecules.

T B. Small nonpolar molecules

F C. Large polar molecules

F D. Large nonpolar molecules

E. Water

16. Which of the following membrane activities does NOT require the expenditure of energy by the cell?

F A. Active transport

T B. Osmosis

F C. Endocytosis

F D. Exocytosis

F E. Synthesis of more membrane

17. The basic unit of contraction is the:

F A. Myosin

F B. Actin

F C. Z-lines

T D. Sarcomeres

18. A motor unit is made up of ________________

F A. All the muscle fibers within a given muscle

T B. Motor neuron and the muscle fibers it innervates

F C. All the neurons going into an individual section of the body

F D. All fascicle and a nerve

19. An ABC transporter:

F A. Is a type of water channel

T B. Uses the energy of ATP to transport solutes.

False F C. Uses gated channels to transport ADP.

D. is present in kidney tubules and prevents dehydration.

E. is a type of porin.

27. Densely staining areas or Nissl bodies of the perikaryon are composed of
F A. Mitochondria.
T B. Rough endoplasmic reticulum.
F C. Golgi apparatus.
F D. Microfilaments and microtubules.

28. A grouping of cell bodies located within the central nervous system is known as a
F A. Tract.
F B. Nerve.
F C. Nucleus.
T D. Ganglion.

29. Myelin sheaths around axons located within the CNS are formed by
F A. Schwann cells.
F B. Microglia.
F C. Astrocytes.
T D. Oligodendrocytes.

30. Which of the following cell types is not a supporting cell or glial cell of the nervous system?
F A. Schwann cell
F B. Oligodendrocyte
F C. Astrocyte
T D. Association neuron

31. The glial cells in the CNS, which are capable of phagocytosis are
F A. Astrocytes.
F B. Oligodendrocytes.
F C. Satellite cells.
T D. Microglia.

32. Within the central nervous system astrocytes are responsible for
F A. Myelinating axons.
F B. Producing cerebrospinal fluid.
F C. Phagocytizing particulate matter
T D. Aid in synapse formation.

33. The specialized cells of the PNS that are surrounded by a basement membrane that is capable of forming a regeneration tube through which a severed peripheral axon can regrow are the
F A. Schwann cells.
F B. Oligodendrocytes.
F C. Microglia.
T D. Astrocytes.

34. The term "voltage regulated" means that the membrane
T A. Gated ion channels open and close with changes in the membrane potential.
T B. Potential is controlled by the Na+/K+ pumps.
F C. Gates will not respond unless the voltage is "regular".
F D. Potential can only be controlled by an oscilloscope.

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42. Curare reduces the size of end plate potentials on the membrane of muscle fibers by
T (A) Competing with acetylcholine for attachment to the postsynaptic receptor proteins. ✓
F B. Blocking the release of acetylcholine from presynaptic vesicles.
F C. Enhancing the breakdown of acetylcholine by acetylcholinesterase.
T D. Blocking the flow of Na+ through open ion channels.

43. The first voltage-regulated gates encountered along the neuron membrane, which initiate the formation of action potentials, are located on the neuron near the
F A. Postsynaptic membrane of the dendrite.
F B. Cell body.
T C. Axon hillock. ✓
F D. Axon terminal.

44. Myelination of axons within the central nervous system would not occur if which of the following cells were not functional?
F A. Astrocytes.
F B. Microglia.
T C. Oligodendrocytes. ✓
F D. Schwann cells.

45. The blood-brain barrier would be compromised by damaging
T A. Astrocytes. ✓
F B. Ependymal cells.
F C. Microglia.
F D. Oligodendrocytes.

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SECTION B: Short Essay
Answer ALL Questions

1. State five (05) physiological importances of the Na+/K+ ATPase (5marks)

- Contributing to the ~~st~~ establishment of resting membrane potential
- Controls the volume of the cell
- ~~It pumps out $3Na^+$ and pumps in $2K^+$~~
- It establishes the electrical gradients for both Na^+ and K^+ in the ECF and ICF
- It ~~s~~ is electrogenic
- It regulates basal metabolic rate.

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2. Describe secondary active transport using an example (5 marks)

- secondary transport is the transport of sodium along with another substance across the membrane
- When sodium is transport inside the cell with another substance in the same direction is called Contrasport.
eg. Na/glucose contrasport
- When sodium is transported inside the cell with another substance in the different or opposite direction is called Counter transport
eg ~~Na/H~~ counter transport H^+/K^+
- The energy released in the movement of sodium is used to transport another substance.

3. State five (05) functions of membrane proteins (5 marks)

- They form channel proteins
- They form carriers proteins
- They participate in cell to cell interaction
- They combine with carbohydrates to form receptors

03

- They ~~act~~ act as cell markers for recognition
- They provide structural support to the cell membrane.

4. In a tabular form differentiate between Red (Type I) and White (Type II) skeletal muscle fibres (5 marks).

- Red muscle fibers have a long ~~chronaxie~~ latent period while ~~white~~ muscle fibers have a short talent period.
- Red muscle fibers respond slowly while white muscle fibers respond quickly (rapidly)
- Red muscle fibers are found in ~~more~~ sustained gross movement while white muscle fibers are found in precise body structures eg hands.
- Red muscle fibers ~~slow~~ are not easily fatigued white white muscle fibers are easily fatigued
- Red muscle fibers are called slow oxidative fibers whil white muscle fibers called fast glycolytic fiber

SECTION C: Long Essay

Answer Question ONE and any other (TWO) questions

1. Draw a well labeled diagram of an action potential of a neuron and state the ionic basis for each phase (10 marks)
2. Draw a well labeled diagram of the neuromuscular junction and describe the excitation contraction coupling mechanism in skeletal muscles (10 marks)
3. Discuss transport across the cell membrane (10 marks)
4. Describe a simple muscle twitch. Add a note on effects of muscle denervation (10 marks)

SECTION B

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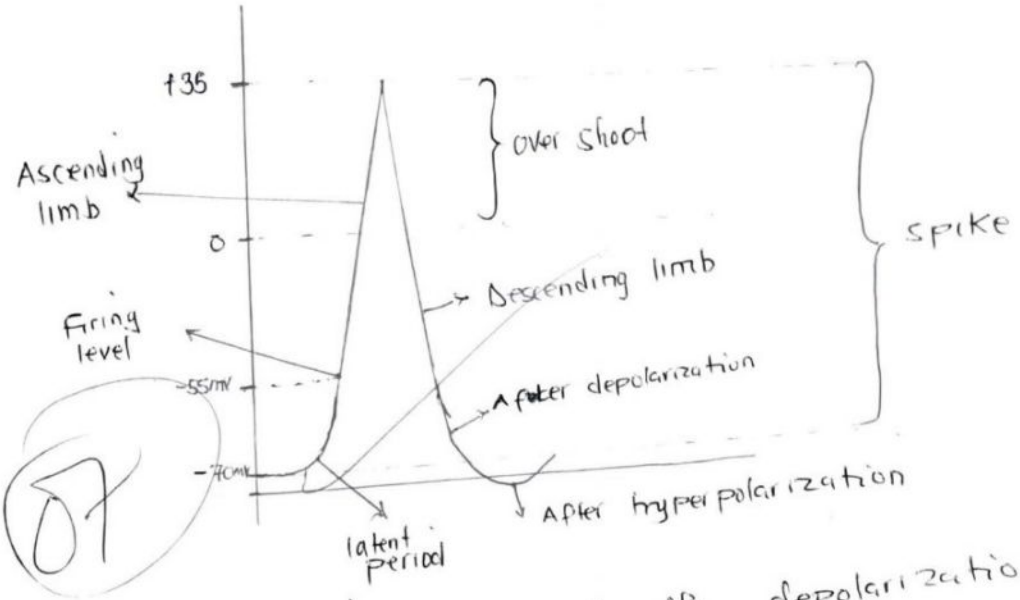
QUESTION ONE



① The ascending limb represents the depolarization phase

- Depolarization is the loss of resting membrane potential

- When a nerve is stimulated the resting membrane potential starts to move from -70mV toward zero because stimulating current are cathodic in nature thereby adding negatives to the outside

- When depolarization reaches 7mV this causes voltage gated sodium channels to open only a few activation gates of sodium open causing Na^+ influx

- When depolarization reaches 15mV that is threshold at -55mV this causes all activation gates of sodium which are outside to open, hence increase in conductance across the membrane and more influx of sodium

- Due to more increase in sodium influx resting membrane potential is lost causing overshoot and the membrane potential becomes +35mV.

② Repolarization is represented by the descending limb

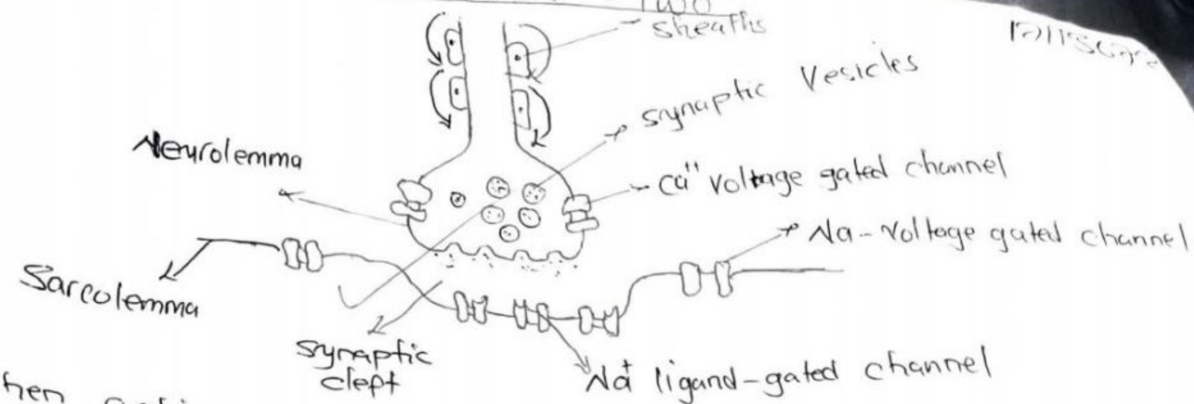
- At overshoot inactivation gates of sodium are closed and activation gates of voltage-gated potassium open causing little ~~outflux~~ efflux of potassium ions because those gates the take long to open and the prolong to close.

- When conductance of potassium ions across the membrane increases there in increase in K^+ ions efflux restoring the membrane potential from +35mV toward -70mV.

- When repolarization is 70 to 80% complete the rate of potassium ions efflux reduces this phase is called After depolarization

- Because activation gates of potassium gates take long to close membrane potential overshoots toward the negative due to more efflux of K^+ ions.

QUESTION TWO



- When action potential reaches the axon terminal it causes calcium voltage gated channels to open causin Ca^{++} influx.
- Calcium influx causes rupture of synaptic vesicles to release acetylcholine an binds to the neurolemma and exocytosis occurs causing acetylcholine to diffuse through the synaptic celeft.
- Acetylcholine binds to the nicotinic receptors on the sodium ligand-gated channel to open causin sodium ions influx in the muscle fiber.
- Acetylcholinesterase enzymes hydrolyze acetylcholine to prevent further excitation.
- Due to sodium influx end plate potential develops at the motor end plate, if it reaches threshold of -70mV action potential is propagated along the sarcolemma.

- Whe action potential reaches the triad in the muscle fiber it causes calcium voltag-gated channel to open and Ca^{++} ions are released in the sarcoplsm.
- Due to increase in Ca^{++} concentration in the sarcoplasm calcium ions bind to Troponin C of the actin filamet, this weaken the interaction of troponic I and actin.
- In the presence of Ca^{++} ions myosin head hydrolyse ATP into ADP and phosphate.
- Active sites on the actin where myosin heads binds are exposed and myosin heads bind to the actin filament causing actin filament to slide between the myosin. hence causing contraction of the muscle
- After sometimes calcium ions are reuptaken into the sarcoplasmic reticulum, whic requires ATP by Ca^{++} pump
- troponin - ~~tropomysim~~ is restored and no interaction between the myosin and the actin and relaxation has occured

07

QUESTION 3

* Transport across plasma membrane is achived due to the structure of the plasma membrane itself.
- plasma membrane has hydrophilic heads exposed to the outer and inner surface of membrane
- Within the membrane there integeral proteins that are able to form channels and carriers that are are involved inthe the transport of substances

- passive transport is the transport of substances down their concetration gradient across the plasmaa ~~plar~~ membrane
  * under passitransport ~~we~~ there
    1. simple diffusion, uses channels, no energy is required hence called down hill

    2. Facilitated difussion no energy is require.

B

Actire transport - tranport aganist concentration gradien
- also called uphill
- uses energy and carrier proteins
  - primary actire transport

  - Secondary actire transpor

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